

T12

$$\alpha = 0,2$$

$$H_0: \{ \sim \frac{1}{3} \{4\} + \frac{1}{6} \{3\} + \frac{1}{4} \{1, 2\} \}$$

$$H_1: \{ \sim \frac{1}{4} \{1, 2, 3, 4\} \}$$

$$p = \frac{L_1}{L_0} = \frac{\left(\frac{1}{4}\right)^2}{\left(\frac{1}{3}\right)^{m_u} \left(\frac{1}{6}\right)^{m_s} \left(\frac{1}{4}\right)^{2-m_u-m_s}} = \frac{\left(\frac{1}{4}\right)^{m_u+m_s}}{\left(\frac{1}{3}\right)^{m_u} \left(\frac{1}{6}\right)^{m_s}}$$

| $(m_3 \ m_4)$       | $(0 \ 0)$     | $(0 \ 1)$     | $(1 \ 0)$     | $(1 \ 1)$     | $(0 \ 2)$      | $(2 \ 0)$      |
|---------------------|---------------|---------------|---------------|---------------|----------------|----------------|
| $p$                 | 1             | $\frac{3}{4}$ | $\frac{3}{2}$ | $\frac{9}{8}$ | $\frac{9}{16}$ | $\frac{9}{4}$  |
| $H_0 \rightarrow p$ | $\frac{1}{4}$ | $\frac{1}{3}$ | $\frac{1}{6}$ | $\frac{1}{9}$ | $\frac{1}{9}$  | $\frac{1}{36}$ |
| $H_1 \rightarrow p$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{4}$ | $\frac{1}{8}$ | $\frac{1}{16}$ | $\frac{1}{16}$ |

$$P(p \geq c | H_0) \leq \alpha = 0,2$$

$$(2 \ 0) \quad p = \frac{1}{36} \leq 0,2$$

$$(1 \ 0) \quad p = \frac{1}{6} + \frac{1}{36} = \frac{7}{36} \leq 0,2$$

$$(1 \ 1) \quad p = \frac{1}{9} + \frac{1}{6} + \frac{1}{36} = \frac{11}{36} > 0,2$$

$$\alpha_1 = \frac{7}{36}$$

G: Выпала хотя бы одна 3

$$w = P(p \geq c | H_1) = \frac{5}{16}$$

$$\alpha_2 = 1 - w = \frac{9}{16}$$